

Composition Pair B1.20 + A2.47: Recursive Governance and Level-Differentiated Authority Distribution as Co-Required Commitments

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its contribution is to formalize the co-required relationship between recursive governance (B1.20) and authority distribution (A2.47) as a composition pair: each commitment requires the other to deliver its intended architectural effect, and neither is complete without the other present and correctly configured.

Abstract

The Coordination Knowledge Substrate (CKS) pattern commits to recursive governance—Paper 1's architectural commitments apply at cell scope, aspect scope, and Self scope independently, with level-appropriate governance at each level (B1.20). It also commits to authority distribution—substrate-resident specification of which human actors hold which governance authority for which governance acts (A2.47). This note formalizes the co-required relationship between these two commitments. B1.20 requires A2.47 to specify who governs at each structural level scope; without level-differentiated authority distribution, recursive governance defaults to deployment-level authority applied uniformly across all levels, defeating the governance isolation that level differentiation was introduced to provide. A2.47 requires B1.20's three structural levels to have meaningful scope dimensions along which authority distribution can be differentiated; without B1.20's level structure, A2.47 distributes authority only at deployment scope, and the distinction between cell-level, aspect-level, and Self-level governance authority does not exist. The pair together enables least-privilege governance and organizational separation of governance concerns. The failure modes of partial co-presence are Authority Ambiguity (B3.26) when A2.47 is absent or uniform, and Deployment-Level-Only Governance (B3.21) when B1.20 is absent.

1. Pair Identification and the Co-Required Relationship

Pair: B1.20 (recursive governance) + A2.47 (authority distribution).

Relationship type: Co-required. Neither commitment delivers its intended architectural effect without the other present and correctly configured. The dependency runs in both directions with no independent direction—this is not an enabling pair where one commitment enables the other, but a pair where each requires the other to be complete.

B1.20 — Recursive governance. The source paper's second claim establishes three structural levels for a CKS Self: cell, aspect, and Self (Paper 2, §5.2). Recursive governance is the commitment that Paper 1's architectural commitments—including human governance, substrate-as-source-of-truth, conflict preservation, path retraceability, and the governance boundary at the instinct/reasoning division—apply at each of these levels independently, not only at the cell level where Paper 1 establishes them. Each level has governance: cell-level governance over cell DNA and action-layer content, aspect-level governance over aspect coordination arrangements and their orchestration substrates, Self-level governance over integration architecture and the composition relationships across aspects. The governance acts available at each level are the same in kind as those Paper 1 defines, instantiated at the appropriate level scope.

A2.47 — Authority distribution. The human-governed commitment (A1.01) establishes that humans retain the rights to inspect, modify, and override substrate content and orchestration rules at all times. Authority distribution specifies the further question: which humans, for which governance acts, at which scope. In a multi-human deployment, multiple humans hold governance authority, but not all humans hold all authority over all substrate content. A2.47 is the substrate-resident specification of this distribution: the record, itself governed substrate content, of who has what governance authority for which governance acts. Authority distribution is not a runtime access-control layer external to the substrate; it is substrate content subject to the same human-governed properties as all other substrate content.

2. B1.20 Requires A2.47: Recursive Governance Without Level-Differentiated Authority Distribution Is Incomplete

Recursive governance per B1.20 means that governance acts occur at three distinct structural levels. A cell operator governs cell-level DNA—the orchestration substrates, conflict-handling rules, and lifecycle policies that define how the cell functions. An aspect coordinator governs aspect-level orchestration—the composition arrangements, routing logic, and purpose-specific coordination substrates that define how the aspect groups its cells. A Self architect governs Self-level integration—the composition relationships across aspects, the institutional substrate content, and the authority architecture under which cells and aspects operate.

These three governance scopes are distinct. The question B1.20's level structure raises but does not by itself answer is: who has authority at each scope? Without A2.47, that question has no substrate-resident answer. The deployment is left with one of two unsatisfactory defaults.

The first default is undifferentiated authority: every actor who holds any governance authority holds it at all levels. An actor authorized to modify cell-level DNA can also modify Self-level integration architecture and aspect-level orchestration substrates. This collapses recursive governance to a flat authority model despite preserving the three-level structural distinction. The isolation that makes recursive governance valuable—that cell operators can govern cells without having authority over Self-level integration decisions, that Self architects can govern integration without needing access to cell-level content—is absent. Any actor can govern at any level, and the governance investment in level differentiation produces no corresponding authority differentiation.

The second default is ungoverned ambiguity: no substrate-resident specification of who governs at each scope exists, and governance acts at each level are either contested or unenforced. Actors at different levels discover at governance-exercise time that their authority boundaries are unclear. A cell operator and a Self architect may both believe they have authority over an aspect-level orchestration substrate that touches both their domains. Neither B1.20's level structure nor the human-governed commitment resolves this without A2.47 specifying the answer.

Both defaults are instances of the Authority Ambiguity anti-pattern (B3.26): governance is recursive in structural principle, but authority is not level-differentiated in operational practice. B2.102 through B2.108 specify per-commitment recursive governance—each Paper 1 commitment applied at each level scope. Each such application requires knowing who has authority to govern that commitment at that level. B2.107, which specifies recursive governance affordances at each level scope, is directly incomplete without A2.47 specifying who holds those affordances at each level. The recursive governance architecture cannot be verified or operated without the corresponding authority distribution.

3. A2.47 Requires B1.20: Authority Distribution Without Recursive Governance Levels Has No Level Scope

Authority distribution per A2.47 specifies who has what governance authority for which governance acts. The operative question is: at which scope? The answer depends on the structural levels the architecture defines.

Without B1.20's recursive governance levels, A2.47 can distribute only what the architecture provides as distributable: governance acts at deployment scope. The distribution is flat—actor A has authority to modify orchestration rules, actor B has authority to inspect substrate content—but there is no cell-level authority distinct from Self-level authority, because without B1.20 there is no cell scope and Self scope as distinct governance domains. Cell-level DNA modification authority, aspect-level coordination governance authority, and Self-level integration architecture authority are distinct categories only because B1.20 makes cell, aspect, and Self distinct governance scopes with distinct governance acts. Without B1.20, those categories do not exist; A2.47's distribution has only one scope dimension, and reduces to the multi-human governance question Paper 1's multi-human axis addresses—who holds authority Paper 1 defines—without the level-differentiation Paper 2 adds.

The Deployment-Level-Only Governance failure mode (B3.21) names this state: A2.47 is present and correctly specified, but only at deployment scope, because B1.20's recursive levels are absent. The deployment correctly distributes authority across actors, but cannot distribute authority *per level* because the level structure is not there to distribute against.

4. Mutual Dependency: What Each Provides to the Other

The co-required relationship has a precise structure: each commitment provides something the other needs to be complete.

B1.20 provides the scope dimensions for A2.47. The three structural levels—cell, aspect, Self—together with recursive governance's commitment that each level has its own governance acts, are the scope dimensions along which A2.47's distribution is differentiated. B2.107's recursive authority architecture specifies governance affordances at each level scope; those affordances are what A2.47 distributes to named actors. Without B1.20's level structure, there are no level-differentiated affordances to distribute.

A2.47 provides the authorization architecture for B1.20's governance acts. Every governance act at every level scope requires an authorized actor. B2.95 cross-level access authority, B2.68 DNA modification authority, and B2.41 birth governance authority are each distributed per A2.47. The human-governed commitment guarantees that the rights to inspect, modify, and override exist; A2.47 specifies which humans hold those rights at which level scope. Without A2.47, governance acts at each level scope are either universally authorized—any actor can govern at any scope, eliminating level isolation—or unauthorized—no substrate-resident specification exists of who can govern what, and governance becomes an exercise of informal organizational authority rather than substrate-resident architectural property.

A2.47 changes under A6.06: changes to authority distribution at any level scope are significant governance events per A6.06's change boundary commitment. Because B1.20 makes governance recursive across three levels, A6.06 applies independently at each level—a change to cell-level authority distribution is a significant event at cell scope, distinct from a Self-level authority change at Self scope.

5. Co-Presence Requirement: What Full and Partial Co-Presence Produce

B1.20 without A2.47—Recursive governance without level-differentiated authority distribution. The three structural levels exist. Governance acts at each level scope are defined. But the specification of who governs at each scope is absent from the substrate, or is present only as flat deployment-scope authority. The result is the Authority Ambiguity anti-pattern (B3.26): governance is architecturally recursive, but any actor with any governance authority can exercise governance at any level. A cell operator can modify Self-level integration architecture. A Self architect can govern cell-level DNA directly. The level isolation that B1.20's recursive structure was introduced to provide is absent in

practice, because A2.47 has not specified the boundaries that would make it real.

This is the more common failure mode in practice, because B1.20's structural levels are typically designed explicitly—the architecture names cell, aspect, and Self as distinct—while A2.47's level-differentiated authority specification is often treated as an organizational rather than architectural matter, deferred to deployment convention rather than substrate content.

A2.47 without B1.20—Authority distribution without recursive governance levels. The authority specification is present and correctly configured, but the level structure it would differentiate against is absent. Actors have clearly specified authority for clearly specified governance acts. But the governance acts exist only at deployment scope; the distinction between governing at cell scope versus aspect scope versus Self scope does not exist architecturally. A2.47's distribution is fully effective at the deployment level and has no further level to differentiate across. This is the Deployment-Level-Only Governance failure mode (B3.21).

Full co-presence—B1.20 and A2.47 both correctly configured. A2.47 specifies authority per level scope per B1.20's three levels. Cell-level DNA modification authority is held by cell operators, not by aspect coordinators or Self architects unless explicitly granted. Aspect-level coordination governance authority is held by aspect coordinators, with cell operators able to exercise it at their cells' participation boundaries per explicit specification. Self-level integration architecture authority is held by Self architects and the organizational humans with appropriate scope. Each governance act at each level scope is traceable to a level-specific A2.47 authorization. B2.107's recursive authority architecture is complete. Governance isolation holds in both principle and operational practice.

6. Operational Implications: Least-Privilege Governance and Separation of Concerns

The co-required pair enables two operational properties that neither commitment produces independently.

Least-privilege governance. With B1.20 and A2.47 both correctly configured, each human actor holds exactly the authority their governance role requires at the level scopes they govern. A cell operator does not require Self-level authority to govern cell-level DNA. A Self architect does not require cell-level access authority to govern integration architecture. An aspect coordinator does not require authority over individual cell internals to govern the aspect's coordination arrangements. The architecture grants authority at the level scope where it is needed and withholds it elsewhere unless explicitly extended. This is the level-differentiated analog of the principle that the human-governed commitment establishes: governance is an authority architecture, not a review workflow, and the authority architecture is specified at the precision the structure provides.

Least-privilege governance permits humans to hold governance roles at the scope they understand and are accountable for without requiring authority over the full deployment. A cell operator governs their domain without being a Self architect; a Self architect governs integration decisions without needing

authority over every cell's internal DNA.

Governance separation of concerns. Different organizational roles govern different levels without full authority over the entire deployment. Cell-level governance and Self-level governance can be organizationally assigned to different teams, with A2.47 carrying those assignments as substrate content. When organizational structure changes, the change is a substrate update under A6.06's change boundary governance—attributed and traceable—rather than an informal organizational handoff. Because A6.06 applies independently at each level under B1.20's recursive structure, a change to cell-level authority distribution is a significant governance event at cell scope, distinct from a Self-level authority change at Self scope.

Cross-level access authority (B2.95) is a specific instance of A2.47 authority: the authority for one level to access content at another level is explicitly specified, not a gap in level isolation. A Self architect who accesses a cell's content directly—which Paper 2 §5.2 permits when purpose requires—does so under explicitly distributed authority, not by bypassing the authority architecture.

7. Detection and Failure Modes

Detection—verifying that the pair is correctly co-present.

B2.107 recursive authority architecture test. For each of the three structural levels—cell, aspect, Self—is A2.47 configured with level-specific authority? Can governance acts at cell scope be traced to cell-scope authorization distinct from aspect-scope or Self-scope authorization? If not at any level, the pair is not fully co-present.

A2.47 authority specification audit. Does the substrate carry a level-differentiated authority specification, or only a deployment-scope specification? A flat specification that names actors and governance acts without level-scope differentiation indicates that A2.47 has not been configured against B1.20's level structure.

Pair completeness test. For each level scope, can a governance act—DNA modification, orchestration substrate change, birth or death governance, cross-level access—be traced to a level-specific A2.47 authorization held by a named actor or role? A deployment unable to complete this trace at any level scope has a partial co-presence failure.

Failure mode 1—Authority Ambiguity (B3.26). A2.47 is absent or configured as a flat deployment-scope specification. B1.20's recursive governance is present—three levels are defined, governance acts at each level are specified—but the authority architecture does not differentiate by level. Any actor with deployment-scope governance authority can govern at any level. Level isolation holds structurally but not in authority practice.

Failure mode 2—Deployment-Level-Only Governance (B3.21). B1.20 is absent—the deployment has not instantiated three recursive structural levels, or has not carried the recursive governance commitment. A2.47 is present and correctly specified, but only at deployment scope. Authority is

correctly distributed among actors, but the distribution is flat: all governance acts are at the same scope regardless of what structural level they would operate at, because the structural levels do not exist as distinct governance domains.

8. Conclusion

B1.20 and A2.47 are co-required: recursive governance requires level-differentiated authority distribution to specify who governs at each structural level scope, and authority distribution requires recursive governance's level structure to have meaningful scope dimensions to differentiate against. A deployment that instantiates B1.20 without A2.47 produces Authority Ambiguity—recursive in structure, flat in authority practice. A deployment that instantiates A2.47 without B1.20 produces Deployment-Level-Only Governance—correctly distributed among actors, but scoped only to the single deployment level.

Full co-presence delivers least-privilege governance, where each human actor holds exactly the authority their level-scope governance role requires; and governance separation of concerns, where different organizational actors govern different structural levels with substrate-resident authority specifications subject to the same governance properties as all other substrate content.

The pair formalizes the intersection at which Paper 2's three-level architecture meets the authority architecture that makes it operationally governed. Both commitments are required; together they are sufficient for level-differentiated governance under the recursive structure Paper 2 establishes.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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